Analysis part2

Dr Giulia Nicole Baldrighi

Turku PET Centre, Clinical Neurosciences

updated: 2025-04-29

Some nice functions in R

Tidyverse library

- summarise data using: group_by(), summarise(), and mutate();
- reshape data between the wide and long formats: pivot_wider() and pivot_longer();
- select() columns and arrange() (sort) rows.

```
library(tidyverse)
```

Melanoma dataset

```
library(readxl)

melanoma <- read_excel("C:/Users/gnbal/Desktop/Lecture/CorsoR_Cadella
```

```
colnames(melanoma)
```

```
## [1] "time"      "status"    "sex"       "age"       "year"      "thickness
## [7] "ulcer"
```

```
head(melanoma)
```

```
## # A tibble: 6 × 7
##    time status   sex   age  year thickness ulcer
##   <dbl>  <dbl> <dbl> <dbl> <dbl>     <dbl> <dbl>
## 1    10      3     1    76  1972      6.76     1
## 2    30      3     1    56  1968      0.65     0
## 3    35      2     1    41  1977      1.34     0
## 4    99      3     0    71  1968      2.9      0
## 5   185      1     1    52  1965     12.1      1
## 6   204      1     1    28  1971      4.84     1
```

group_by(), summarise()

To use *group_by*0, we need to recode variables with classes represented by numbers into classes coded as factors

```
melanoma$sex <- as.factor(melanoma$sex)

levels(melanoma$sex)
```

```
## [1] "0" "1"
```

```
melanoma %>%
  group_by(sex) %>%
  summarise(mean_thickness = mean(thickness, na.rm = TRUE))
```

```
## # A tibble: 2 × 2
##   sex   mean_thickness
##   <fct>          <dbl>
## 1 0               2.49
## 2 1               3.61
```

mutate()

To transform the thickness from mm^3 to cm^3

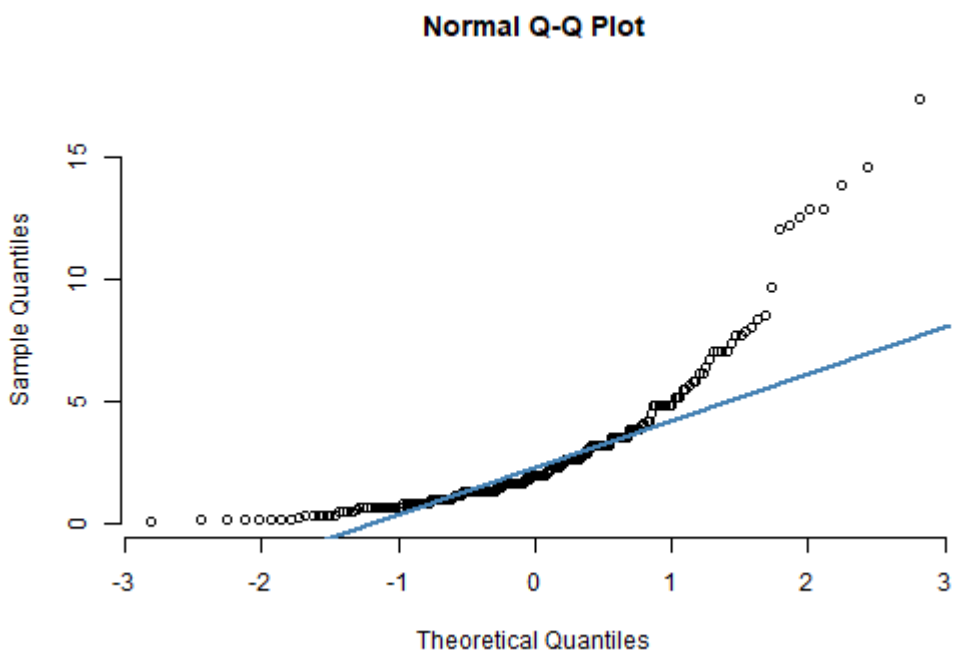
```
melanoma %>%
  mutate(thickness2 = thickness / 1000)
```

```
## # A tibble: 205 × 8
##     time status sex     age  year thickness ulcer thickness2
##    <dbl>  <dbl> <fct> <dbl> <dbl>     <dbl> <dbl>      <dbl>
##  1    10      3 1        76  1972      6.76     1    0.00676
##  2    30      3 1        56  1968      0.65     0    0.00065
##  3    35      2 1        41  1977      1.34     0    0.00134
##  4    99      3 0        71  1968      2.9      0    0.0029
##  5   185      1 1        52  1965     12.1      1    0.0121
##  6   204      1 1        28  1971      4.84     1    0.00484
##  7   210      1 1        77  1972      5.16     1    0.00516
##  8   232      3 0        60  1974      3.22     1    0.00322
##  9   232      1 1        49  1968     12.9      1    0.0129
## 10   279      1 0        68  1971      7.41     1    0.00741
## # ℹ 195 more rows
```

Numeric variables

Exploring variables' distribution

```
qqnorm(melanoma$thickness, pch = 1, frame = FALSE)
qqline(melanoma$thickness, col = "steelblue", lwd = 2)
```



Testing for distribution

Normality

```r
shapiro.test(melanoma$thickness)
```

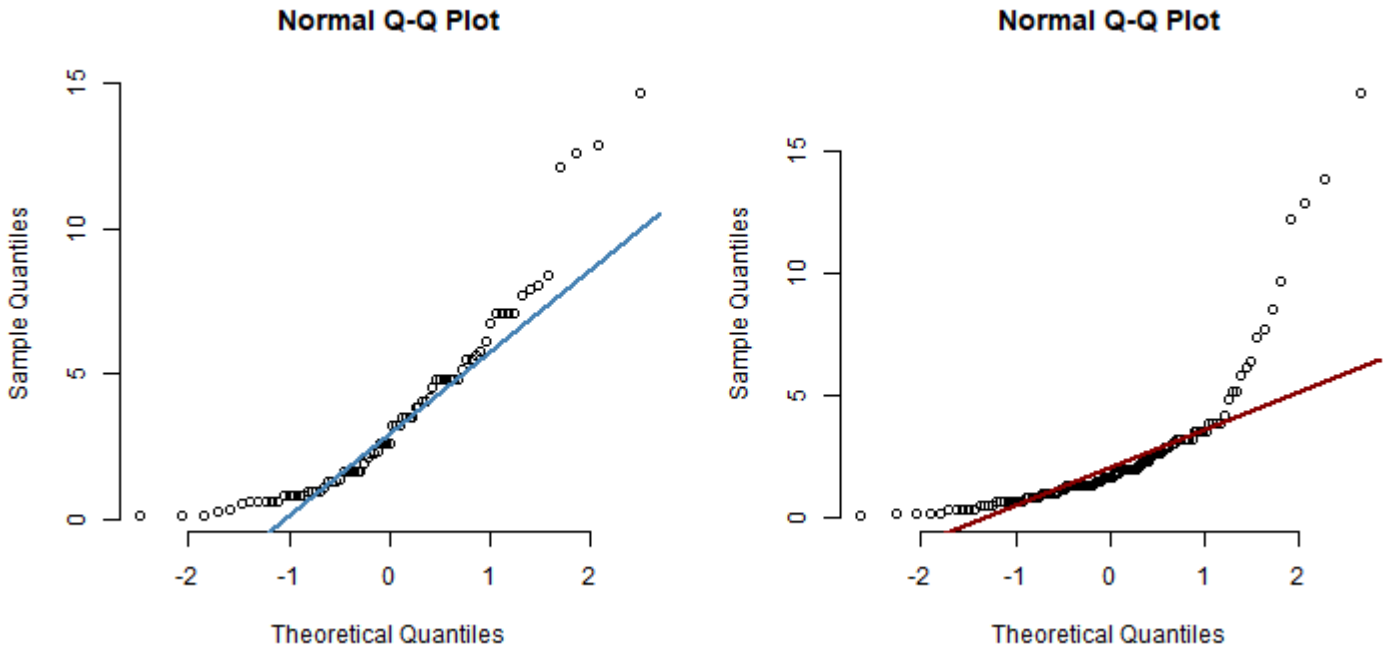
```
##
##     Shapiro-Wilk normality test
##
## data:  melanoma$thickness
## W = 0.76818, p-value < 2.2e-16
```

Skewness

```r
library(statpsych)
test.skew(melanoma$thickness)
```

```
##  Skewness p
##    2.1655 0
```

```
par(mfrow=c(1,2))
qqnorm(melanoma$thickness[melanoma$sex=="1"], pch = 1, frame = FALSE)
qqline(melanoma$thickness[melanoma$sex=="1"], col = "steelblue", lwd
qqnorm(melanoma$thickness[melanoma$sex=="0"], pch = 1, frame = FALSE)
qqline(melanoma$thickness[melanoma$sex=="0"], col = "darkred", lwd =
```



Testing for distribution between two groups

Normality

```
shapiro.test(melanoma$thickness[melanoma$sex=="1"])
```

```
##
##     Shapiro-Wilk normality test
##
## data:  melanoma$thickness[melanoma$sex == "1"]
## W = 0.86274, p-value = 5.077e-07
```

```
shapiro.test(melanoma$thickness[melanoma$sex=="0"])
```

```
##
##     Shapiro-Wilk normality test
##
## data:  melanoma$thickness[melanoma$sex == "0"]
## W = 0.67621, p-value = 2.687e-15
```

Testing for distribution between two groups

Skewness

```
test.skew(melanoma$thickness[melanoma$sex=="1"])
```

```
##  Skewness p
##    1.3986 0
```

```
test.skew(melanoma$thickness[melanoma$sex=="0"])
```

```
##  Skewness p
##    2.9291 0
```

Testing for homoschedasticy between two groups

Homoschedasticity is the homogeneity of variances of the numeric variable within the two groups that we are comparing

```
library(car)

melanoma$sex <- as.factor(melanoma$sex)

leveneTest(thickness ~ sex, data = melanoma)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   1   5.635 0.01854 *
##       203
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assumptions of parametric test (homoschedasticity+normality)

- We check homogeneity of variances (equal variances) because standard t-test assumes it
- If the variances in the two groups are very different, the classic t-test becomes unreliable
- It can give wrong p-values (too small or too large)
- You might falsely think there is a difference when there is not (Type I error)
- Or miss a real difference (Type II error)

If assumptions of parametric test are not respected

- In a t-test, the standard error (which controls the test statistic) depends on the assumption that both groups have similar variability
- Welch's test adjusts for different variances and different sample sizes — it’s more reliable when variances are unequal

Test to use for comparing two groups

Scenarios

- Normality OK, equal variances **t-test** (Student's t-test)
- Normality OK, unequal variances **Welch's t-test** (var.equal = FALSE)
- Normality NOT OK, any variances **Wilcoxon rank-sum test**

Categorical variables

```
melanoma$status <- as.factor(melanoma$status)
table(melanoma$sex, melanoma$status)
```

```
##
##      1  2  3
##   0 28 91  7
##   1 29 43  7
```

- **Chi-square** test is used to analyze relationships between categorical variables
- Specifically, you'd use it when you want to determine if there's a statistically significant association between two or more categorical variables

Testing for independence

Assumptions

- Step 1: Determine whether both variables are categorical
- Step 2: Determine whether simple random sampling was applied
- Step 3: Determine whether all expected frequencies are greater than or equal to 1

```
chisq.test(table(melanoma$sex, melanoma$status))
```

```
##
##     Pearson's Chi-squared test
##
## data:  table(melanoma$sex, melanoma$status)
## X-squared = 6.793, df = 2, p-value = 0.03349
```